simplification

April 24, 2026

[28]:
```
import warnings
warnings.filterwarnings("ignore")

from pathlib import Path
import subprocess
import sys
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    roc_auc_score,
    average_precision_score,
    brier_score_loss,
 )
from sklearn.calibration import CalibratedClassifierCV, calibration_curve
from sklearn.isotonic import IsotonicRegression

try:
    from lifelines import KaplanMeierFitter, CoxPHFitter
    from lifelines.utils import concordance_index
except ModuleNotFoundError:
    print("Installing lifelines...")
    subprocess.check_call([sys.executable, "-m", "pip", "install", "lifelines",␣
 ↪"-q"])

    from lifelines import KaplanMeierFitter, CoxPHFitter
    from lifelines.utils import concordance_index
```

[29]:
```
RANDOM_STATE = 42
```

```python
# Turn sampling on only if runtime becomes annoying on your machine.
USE_SAMPLE = False
SAMPLE_SIZE = 250_000

SIX_HORIZON = 6
SHORT_HORIZON = 12
EIGHTEEN_HORIZON = 18
LONG_HORIZON = 24

TRAIN_END = "2014-12-31"
VALID_START, VALID_END = "2015-01-01", "2015-12-31"
TEST_START, TEST_END = "2016-01-01", "2019-12-31"

DATA_DIR = Path("data")
RAW_DIR = DATA_DIR / "raw"
OUTPUT_DIR = DATA_DIR / "simplification_PD"
OUTPUT_DIR.mkdir(parents=True, exist_ok=True)

print("Output directory:", OUTPUT_DIR.resolve())
```

```
Output directory:
/Users/pattonpatton/Desktop/RiskProject/Risk_Management_Project/Time
Based/data/simplification_PD
```

0.1 2. Find and load the LendingClub file

[30]:
```python
def find_accepted_file():
    candidates = [
        Path("data/raw/lendingclub/accepted_2007_to_2018Q4.csv.gz"),
        Path("data/raw/accepted_2007_to_2018Q4.csv.gz"),
        Path("accepted_2007_to_2018Q4.csv.gz"),
        Path("/mnt/data/accepted_2007_to_2018Q4.csv.gz"),
        Path("/mnt/data/data/raw/lendingclub/accepted_2007_to_2018Q4.csv.gz"),
    ]
    for path in candidates:
        if path.exists():
            return path
    raise FileNotFoundError(
        "Could not find accepted_2007_to_2018Q4.csv.gz. Put it in data/raw/
↪lendingclub/ "
        "or edit find_accepted_file()."
    )

accepted_path = find_accepted_file()
print("Using raw file:", accepted_path)
```

```
Using raw file: data/raw/lendingclub/accepted_2007_to_2018Q4.csv.gz
```

[31]:
```
key_columns = [
    "id",
    "loan_status",
    "issue_d",
    "term",
    "grade",
    "int_rate",
    "loan_amnt",
    "installment",
    "emp_length",
    "home_ownership",
    "annual_inc",
    "verification_status",
    "purpose",
    "addr_state",
    "dti",
    "delinq_2yrs",
    "inq_last_6mths",
    "open_acc",
    "pub_rec",
    "revol_bal",
    "revol_util",
    "total_acc",
    "mort_acc",
    "pub_rec_bankruptcies",
    "fico_range_low",
    "fico_range_high",
    "earliest_cr_line",
    "last_pymnt_d",
    "last_credit_pull_d",
]

df_raw = pd.read_csv(
    accepted_path,
    usecols=lambda c: c in key_columns,
    low_memory=False,
    compression="gzip",
)

if USE_SAMPLE:
    df_raw = df_raw.sample(min(SAMPLE_SIZE, len(df_raw)),␣
 ↪random_state=RANDOM_STATE).copy()

print("Raw shape:", df_raw.shape)
display(df_raw.head(3))
```

```
Raw shape: (2260701, 29)
```

```
         id  loan_amnt        term  int_rate  installment grade emp_length  \
0  68407277     3600.0   36 months     13.99       123.03     C  10+ years
1  68355089    24700.0   36 months     11.99       820.28     C  10+ years
2  68341763    20000.0   60 months     10.78       432.66     B  10+ years

  home_ownership  annual_inc verification_status  … inq_last_6mths open_acc  \
0       MORTGAGE     55000.0        Not Verified  …            1.0      7.0
1       MORTGAGE     65000.0        Not Verified  …            4.0     22.0
2       MORTGAGE     63000.0        Not Verified  …            0.0      6.0

  pub_rec revol_bal  revol_util  total_acc last_pymnt_d  last_credit_pull_d  \
0     0.0    2765.0        29.7       13.0     Jan-2019            Mar-2019
1     0.0   21470.0        19.2       38.0     Jun-2016            Mar-2019
2     0.0    7869.0        56.2       18.0     Jun-2017            Mar-2019

   mort_acc  pub_rec_bankruptcies
0       1.0                   0.0
1       4.0                   0.0
2       5.0                   0.0

[3 rows x 29 columns]
```

0.2 3. Clean the loan-level data

[32]:
```
df = df_raw.copy()

for c in ["issue_d", "earliest_cr_line", "last_pymnt_d", "last_credit_pull_d"]:
    df[c] = pd.to_datetime(df[c], format="%b-%Y", errors="coerce")

status = df["loan_status"].fillna("").astype(str)

# Keep the default definition simple and consistent.
default_mask = status.str.contains("Charged Off|Default", case=False,␣
 ↪regex=True)
payoff_mask = status.str.contains("Fully Paid", case=False, regex=False) &␣
 ↪(~default_mask)

df["default_event"] = default_mask.astype(int)
df["payoff_event"] = payoff_mask.astype(int)
df["event_type"] = np.select(
    [df["default_event"] == 1, df["payoff_event"] == 1],
    [1, 2],
    default=0,
)

df["term_num"] = (
    df["term"].astype(str).str.extract(r"(\d+)", expand=False).astype(float)
```

```
)

# Strip percentage signs where needed.
df["int_rate"] = (
    df["int_rate"].astype(str).str.replace("%", "", regex=False)
)
df["int_rate"] = pd.to_numeric(df["int_rate"], errors="coerce")

df["fico_avg"] = (df["fico_range_low"] + df["fico_range_high"]) / 2
df["credit_history_years"] = (
    (df["issue_d"] - df["earliest_cr_line"]).dt.days / 365.25
)

# Origination-safe engineered features
income_denom = df["annual_inc"].replace(0, np.nan)
df["loan_to_income"] = df["loan_amnt"] / income_denom
df["installment_to_income"] = df["installment"] / income_denom
df["inq_to_total_acc"] = df["inq_last_6mths"] / (df["total_acc"].fillna(0) + 1)
df["delinq_flag"] = (df["delinq_2yrs"].fillna(0) > 0).astype(int)
df["fico_x_dti"] = df["fico_avg"] * df["dti"]
df[["loan_to_income", "installment_to_income", "inq_to_total_acc",␣
 ↪"fico_x_dti"]] = df[["loan_to_income", "installment_to_income",␣
 ↪"inq_to_total_acc", "fico_x_dti"]].replace([np.inf, -np.inf], np.nan)

df["end_date"] = df["last_pymnt_d"]
missing_end = df["end_date"].isna()
df.loc[missing_end, "end_date"] = df.loc[missing_end, "last_credit_pull_d"]

def month_diff(start, end):
    return (end.dt.year - start.dt.year) * 12 + (end.dt.month - start.dt.month)

df = df[df["issue_d"].notna()].copy()
df = df[df["end_date"].notna()].copy()

df["duration_months"] = month_diff(df["issue_d"], df["end_date"]).clip(lower=1)

numeric_cols = [
    "int_rate",
    "loan_amnt",
    "installment",
    "annual_inc",
    "dti",
    "delinq_2yrs",
    "inq_last_6mths",
    "open_acc",
    "pub_rec",
    "revol_bal",
```

```
    "revol_util",
    "total_acc",
    "mort_acc",
    "pub_rec_bankruptcies",
    "fico_avg",
    "credit_history_years",
    "term_num",
    "loan_to_income",
    "installment_to_income",
    "inq_to_total_acc",
    "delinq_flag",
    "fico_x_dti",
]

for c in numeric_cols:
    df[c] = pd.to_numeric(df[c], errors="coerce")

df_model = df[df["issue_d"] <= pd.Timestamp(TEST_END)].copy()

print("Clean modeling shape:", df_model.shape)
print("Default share:", round(df_model["default_event"].mean(), 4))
print("Payoff share :", round(df_model["payoff_event"].mean(), 4))
display(df_model[["id", "issue_d", "loan_status", "default_event",␣
 ↪"payoff_event", "duration_months"]].head())
```

```
Clean modeling shape: (2260668, 42)
Default share: 0.1192
Payoff share : 0.4772
```

```
         id    issue_d loan_status  default_event  payoff_event  \
0  68407277 2015-12-01  Fully Paid              0             1
1  68355089 2015-12-01  Fully Paid              0             1
2  68341763 2015-12-01  Fully Paid              0             1
3  66310712 2015-12-01     Current              0             0
4  68476807 2015-12-01  Fully Paid              0             1

   duration_months
0               37
1                6
2               18
3               38
4                7
```

[33]:
```
train_loans = df_model[df_model["issue_d"] <= pd.Timestamp(TRAIN_END)].copy()
valid_loans = df_model[
    (df_model["issue_d"] >= pd.Timestamp(VALID_START)) &
    (df_model["issue_d"] <= pd.Timestamp(VALID_END))
].copy()
```

```
test_loans = df_model[
    (df_model["issue_d"] >= pd.Timestamp(TEST_START)) &
    (df_model["issue_d"] <= pd.Timestamp(TEST_END))
].copy()

split_table = pd.DataFrame([
    {"split": "train", "n_loans": len(train_loans), "issue_start":␣
 ↪train_loans["issue_d"].min(), "issue_end": train_loans["issue_d"].max()},
    {"split": "valid", "n_loans": len(valid_loans), "issue_start":␣
 ↪valid_loans["issue_d"].min(), "issue_end": valid_loans["issue_d"].max()},
    {"split": "test", "n_loans": len(test_loans), "issue_start":␣
 ↪test_loans["issue_d"].min(), "issue_end": test_loans["issue_d"].max()},
])
display(split_table)
```

```
   split  n_loans issue_start  issue_end
0  train   466345  2007-06-01 2014-12-01
1  valid   421095  2015-01-01 2015-12-01
2   test  1373228  2016-01-01 2018-12-01
```

0.3 5. Build fixed-horizon labels

[34]:
```
def make_horizon_label(loans, horizon):
    out = loans.copy()
    out["known_horizon_outcome"] = (
        (out["duration_months"] >= horizon) |
        ((out["event_type"].isin([1, 2])) & (out["duration_months"] <= horizon))
    )
    out = out[out["known_horizon_outcome"]].copy()
    out[f"default_within_{horizon}m"] = (
        (out["default_event"] == 1) & (out["duration_months"] <= horizon)
    ).astype(int)
    return out

train_6 = make_horizon_label(train_loans, SIX_HORIZON)
valid_6 = make_horizon_label(valid_loans, SIX_HORIZON)
test_6 = make_horizon_label(test_loans, SIX_HORIZON)

train_12 = make_horizon_label(train_loans, SHORT_HORIZON)
valid_12 = make_horizon_label(valid_loans, SHORT_HORIZON)
test_12 = make_horizon_label(test_loans, SHORT_HORIZON)

train_18 = make_horizon_label(train_loans, EIGHTEEN_HORIZON)
valid_18 = make_horizon_label(valid_loans, EIGHTEEN_HORIZON)
test_18 = make_horizon_label(test_loans, EIGHTEEN_HORIZON)

train_24 = make_horizon_label(train_loans, LONG_HORIZON)
```

```
valid_24 = make_horizon_label(valid_loans, LONG_HORIZON)
test_24 = make_horizon_label(test_loans, LONG_HORIZON)

label_summary = pd.DataFrame([
    {"dataset": "train_6", "n": len(train_6), "default_rate":␣
 ↪train_6[f"default_within_{SIX_HORIZON}m"].mean()},
    {"dataset": "valid_6", "n": len(valid_6), "default_rate":␣
 ↪valid_6[f"default_within_{SIX_HORIZON}m"].mean()},
    {"dataset": "test_6",  "n": len(test_6),  "default_rate":␣
 ↪test_6[f"default_within_{SIX_HORIZON}m"].mean()},
    {"dataset": "train_12", "n": len(train_12), "default_rate":␣
 ↪train_12[f"default_within_{SHORT_HORIZON}m"].mean()},
    {"dataset": "valid_12", "n": len(valid_12), "default_rate":␣
 ↪valid_12[f"default_within_{SHORT_HORIZON}m"].mean()},
    {"dataset": "test_12",  "n": len(test_12),  "default_rate":␣
 ↪test_12[f"default_within_{SHORT_HORIZON}m"].mean()},
    {"dataset": "train_18", "n": len(train_18), "default_rate":␣
 ↪train_18[f"default_within_{EIGHTEEN_HORIZON}m"].mean()},
    {"dataset": "valid_18", "n": len(valid_18), "default_rate":␣
 ↪valid_18[f"default_within_{EIGHTEEN_HORIZON}m"].mean()},
    {"dataset": "test_18",  "n": len(test_18),  "default_rate":␣
 ↪test_18[f"default_within_{EIGHTEEN_HORIZON}m"].mean()},
    {"dataset": "train_24", "n": len(train_24), "default_rate":␣
 ↪train_24[f"default_within_{LONG_HORIZON}m"].mean()},
    {"dataset": "valid_24", "n": len(valid_24), "default_rate":␣
 ↪valid_24[f"default_within_{LONG_HORIZON}m"].mean()},
    {"dataset": "test_24",  "n": len(test_24),  "default_rate":␣
 ↪test_24[f"default_within_{LONG_HORIZON}m"].mean()},
])
display(label_summary)
```

	dataset	n	default_rate
0	train_6	466345	0.018220
1	valid_6	421095	0.020767
2	test_6	1247606	0.023025
3	train_12	466345	0.051022
4	valid_12	421095	0.060556
5	test_12	1016181	0.065078
6	train_18	466345	0.083374
7	valid_18	421095	0.100763
8	test_18	847087	0.109652
9	train_24	466345	0.113403
10	valid_24	421095	0.133633
11	test_24	707053	0.151501

0.4 6. Features

[35]:
```
model_numeric = [
    "loan_amnt",
    "installment",
    "annual_inc",
    "dti",
    "delinq_2yrs",
    "inq_last_6mths",
    "open_acc",
    "pub_rec",
    "total_acc",
    "mort_acc",
    "pub_rec_bankruptcies",
    "fico_avg",
    "credit_history_years",
    "term_num",
    "loan_to_income",
    "installment_to_income",
    "inq_to_total_acc",
    "delinq_flag",
    "fico_x_dti",
]

model_categorical = [
    "addr_state",
    "purpose",
    "home_ownership",
    "emp_length",
    "verification_status",
]

preprocessor = ColumnTransformer(
    transformers=[
        ("num", Pipeline([
            ("imputer", SimpleImputer(strategy="median")),
            ("scaler", StandardScaler()),
        ]), model_numeric),
        ("cat", Pipeline([
            ("imputer", SimpleImputer(strategy="most_frequent")),
            ("onehot", OneHotEncoder(handle_unknown="ignore")),
        ]), model_categorical),
    ]
)
```

0.5 7. Frequency-based models: LPM and Logit

[36]:
```
def fit_frequency_models(train_df, valid_df, test_df, horizon):
    ycol = f"default_within_{horizon}m"
    X_train = train_df[model_numeric + model_categorical]
    y_train = train_df[ycol]
    X_valid = valid_df[model_numeric + model_categorical]
    y_valid = valid_df[ycol]
    X_test = test_df[model_numeric + model_categorical]
    y_test = test_df[ycol]

    lpm = Pipeline([
        ("prep", preprocessor),
        ("model", LinearRegression()),
    ])
    logit = Pipeline([
        ("prep", preprocessor),
        ("model", LogisticRegression(max_iter=2000, C=0.5,␣
 ↪random_state=RANDOM_STATE)),
    ])
    rf_raw = Pipeline([
        ("prep", preprocessor),
        ("model", RandomForestClassifier(
            n_estimators=200,
            max_depth=12,
            min_samples_leaf=50,
            max_features="sqrt",
            class_weight="balanced_subsample",
            max_samples=0.7,
            random_state=RANDOM_STATE,
            n_jobs=-1,
        )),
    ])
    rf_raw.fit(X_train, y_train)
    rf = CalibratedClassifierCV(estimator=rf_raw, method="sigmoid", cv=3)

    lpm.fit(X_train, y_train)
    logit.fit(X_train, y_train)
    rf.fit(X_train, y_train)

    lpm_valid_raw = lpm.predict(X_valid)
    lpm_test_raw = lpm.predict(X_test)
    lpm_valid = np.clip(lpm_valid_raw, 0, 1)
    lpm_test = np.clip(lpm_test_raw, 0, 1)

    logit_valid = logit.predict_proba(X_valid)[:, 1]
    logit_test = logit.predict_proba(X_test)[:, 1]
```

```
rf_valid = rf.predict_proba(X_valid)[:, 1]
rf_test = rf.predict_proba(X_test)[:, 1]

results = pd.DataFrame([
    {
        "model": f"LPM_{horizon}m",
        "valid_roc_auc": roc_auc_score(y_valid, lpm_valid),
        "valid_pr_auc": average_precision_score(y_valid, lpm_valid),
        "valid_brier": brier_score_loss(y_valid, lpm_valid),
        "test_roc_auc": roc_auc_score(y_test, lpm_test),
        "test_pr_auc": average_precision_score(y_test, lpm_test),
        "test_brier": brier_score_loss(y_test, lpm_test),
    },
    {
        "model": f"Logit_{horizon}m",
        "valid_roc_auc": roc_auc_score(y_valid, logit_valid),
        "valid_pr_auc": average_precision_score(y_valid, logit_valid),
        "valid_brier": brier_score_loss(y_valid, logit_valid),
        "test_roc_auc": roc_auc_score(y_test, logit_test),
        "test_pr_auc": average_precision_score(y_test, logit_test),
        "test_brier": brier_score_loss(y_test, logit_test),
    },
    {
        "model": f"RF_{horizon}m",
        "valid_roc_auc": roc_auc_score(y_valid, rf_valid),
        "valid_pr_auc": average_precision_score(y_valid, rf_valid),
        "valid_brier": brier_score_loss(y_valid, rf_valid),
        "test_roc_auc": roc_auc_score(y_test, rf_test),
        "test_pr_auc": average_precision_score(y_test, rf_test),
        "test_brier": brier_score_loss(y_test, rf_test),
    },
])

payload = {
    "lpm": lpm,
    "logit": logit,
    "rf": rf,
    "rf_raw": rf_raw,
    "valid_pred_lpm": lpm_valid,
    "test_pred_lpm": lpm_test,
    "valid_pred_logit": logit_valid,
    "test_pred_logit": logit_test,
    "valid_pred_rf": rf_valid,
    "test_pred_rf": rf_test,
    "y_valid": y_valid,
    "y_test": y_test,
```

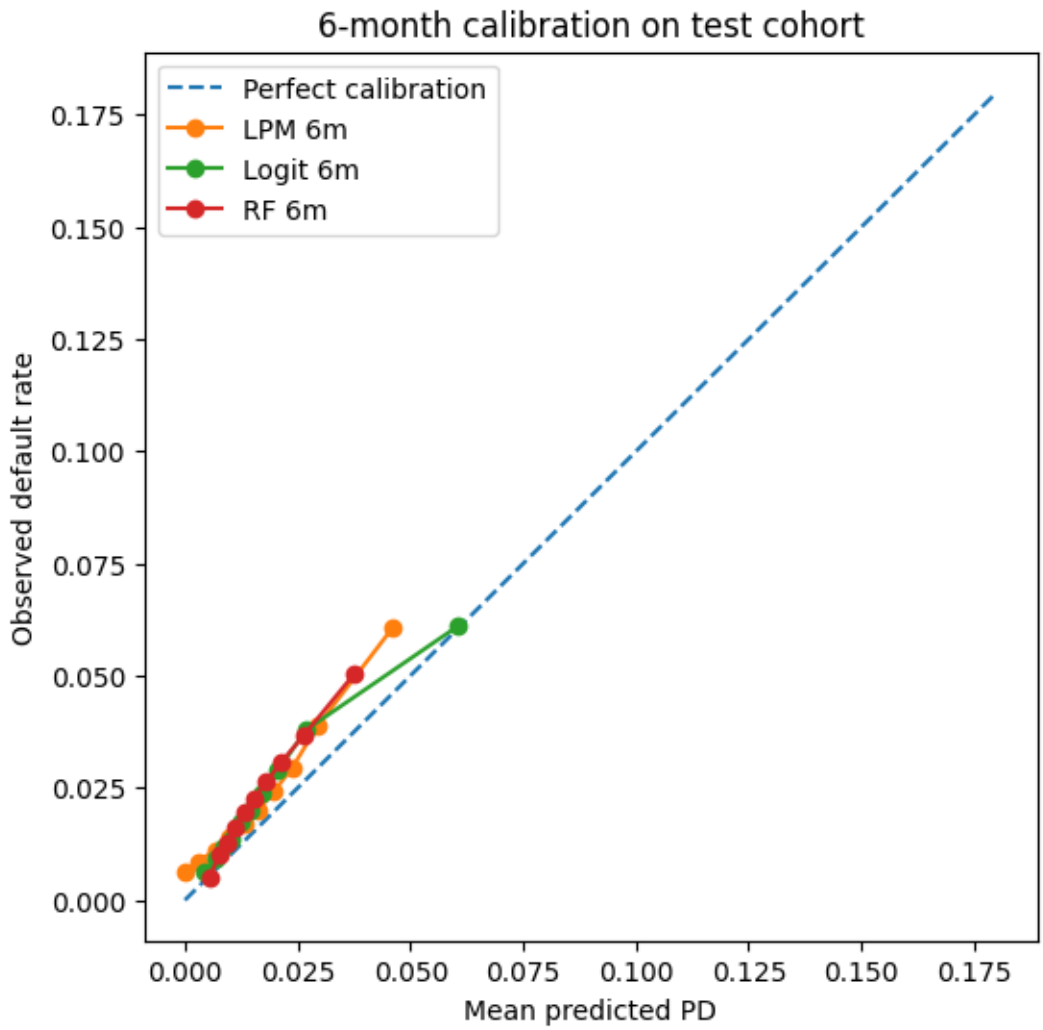
```
        "results": results,
    }
    return payload

freq6 = fit_frequency_models(train_6, valid_6, test_6, SIX_HORIZON)
freq12 = fit_frequency_models(train_12, valid_12, test_12, SHORT_HORIZON)
freq18 = fit_frequency_models(train_18, valid_18, test_18, EIGHTEEN_HORIZON)
freq24 = fit_frequency_models(train_24, valid_24, test_24, LONG_HORIZON)

frequency_results = pd.concat(
    [
        freq6["results"].assign(horizon_months=SIX_HORIZON),
        freq12["results"].assign(horizon_months=SHORT_HORIZON),
        freq18["results"].assign(horizon_months=EIGHTEEN_HORIZON),
        freq24["results"].assign(horizon_months=LONG_HORIZON),
    ],
    ignore_index=True,
)

display(frequency_results)
```

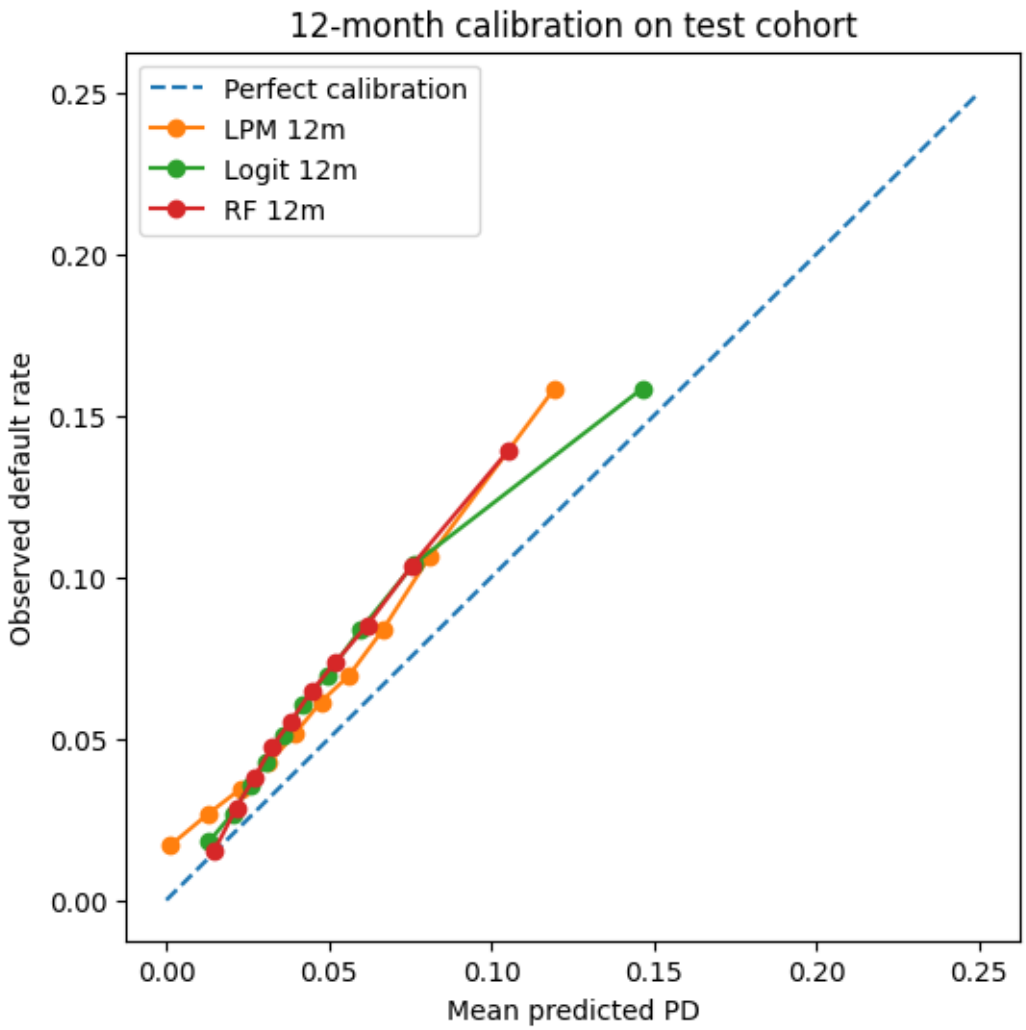
```
        model  valid_roc_auc  valid_pr_auc  valid_brier  test_roc_auc  \
0      LPM_6m       0.698025      0.043929     0.020149      0.686291
1    Logit_6m       0.697541      0.046132     0.020145      0.684235
2       RF_6m       0.672954      0.039203     0.020192      0.662875
3     LPM_12m       0.691000      0.122241     0.055564      0.680430
4   Logit_12m       0.689929      0.122790     0.055531      0.677830
5      RF_12m       0.673264      0.111603     0.055767      0.663990
6     LPM_18m       0.683641      0.185799     0.087621      0.677735
7   Logit_18m       0.682818      0.187610     0.087569      0.675847
8      RF_18m       0.670218      0.176071     0.087967      0.666832
9     LPM_24m       0.681134      0.234117     0.110959      0.680005
10  Logit_24m       0.680637      0.238374     0.110807      0.679218
11     RF_24m       0.669231      0.226967     0.111383      0.671190

    test_pr_auc  test_brier  horizon_months
0      0.047357    0.022415               6
1      0.048753    0.023031               6
2      0.042169    0.022374               6
3      0.125413    0.059860              12
4      0.125274    0.060490              12
5      0.116345    0.059938              12
6      0.197494    0.095094              18
7      0.198873    0.095529              18
8      0.190046    0.095451              18
9      0.261506    0.124162              24
10     0.266726    0.124262              24
11     0.256947    0.124861              24
```

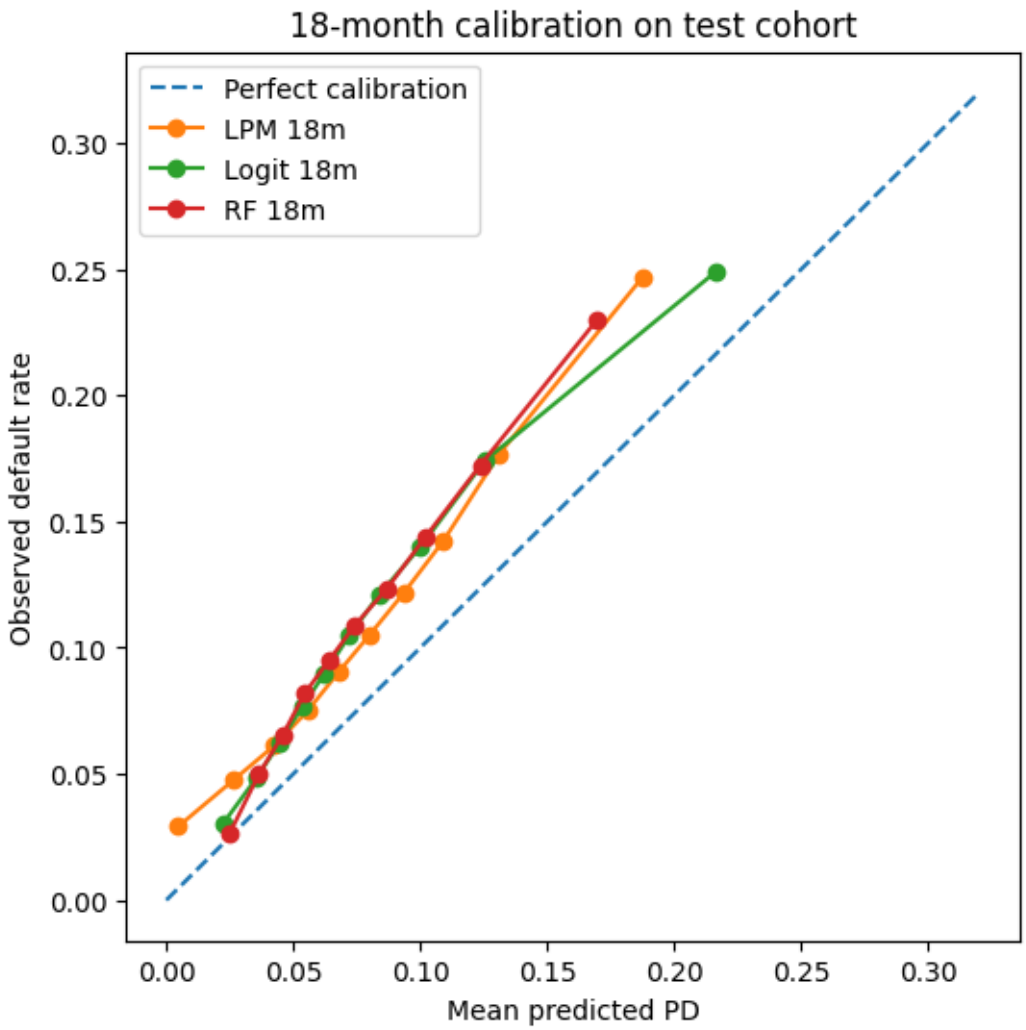
0.5.1 7.1 Accuracy visuals for the frequency models

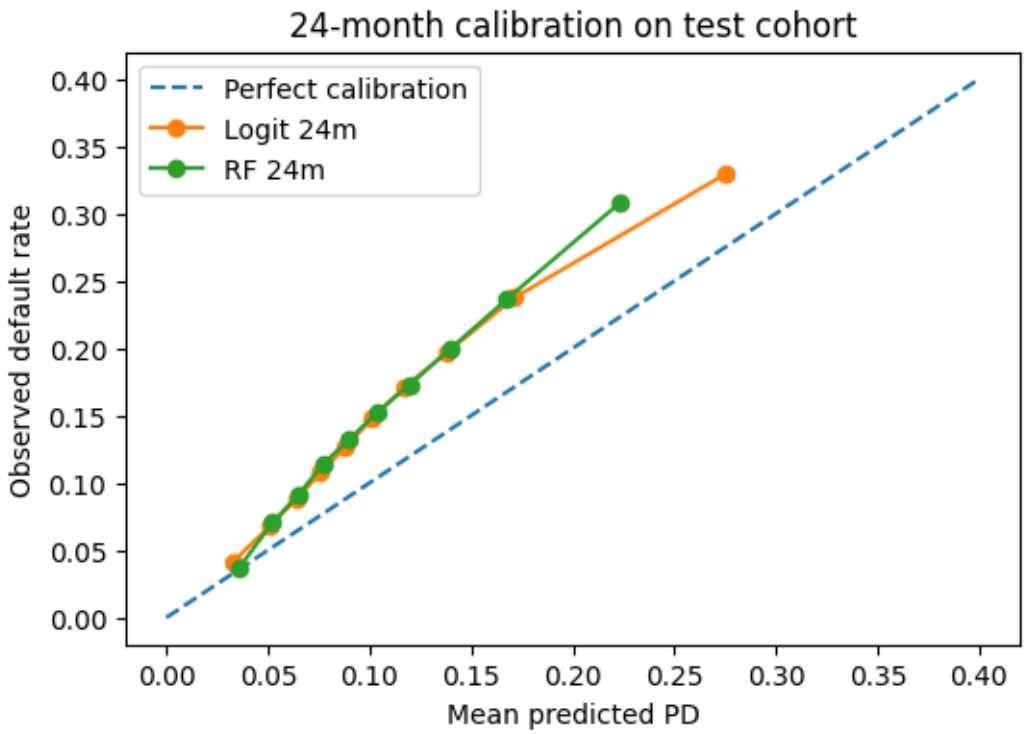
[37]:
```
# 6m calibration: LPM vs Logit vs RF
prob_true_lpm_6, prob_pred_lpm_6 = calibration_curve(freq6["y_test"],␣
 ↪freq6["test_pred_lpm"], n_bins=10, strategy="quantile")
prob_true_logit_6, prob_pred_logit_6 = calibration_curve(freq6["y_test"],␣
 ↪freq6["test_pred_logit"], n_bins=10, strategy="quantile")
prob_true_rf_6, prob_pred_rf_6 = calibration_curve(freq6["y_test"],␣
 ↪freq6["test_pred_rf"], n_bins=10, strategy="quantile")

plt.figure(figsize=(6, 6))
plt.plot([0, 0.18], [0, 0.18], linestyle="--", label="Perfect calibration")
plt.plot(prob_pred_lpm_6, prob_true_lpm_6, marker="o", label="LPM 6m")
plt.plot(prob_pred_logit_6, prob_true_logit_6, marker="o", label="Logit 6m")
plt.plot(prob_pred_rf_6, prob_true_rf_6, marker="o", label="RF 6m")
plt.xlabel("Mean predicted PD")
plt.ylabel("Observed default rate")
plt.title("6-month calibration on test cohort")
plt.legend()
plt.show()

# 12m calibration: LPM vs Logit vs RF
prob_true_lpm_12, prob_pred_lpm_12 = calibration_curve(freq12["y_test"],␣
 ↪freq12["test_pred_lpm"], n_bins=10, strategy="quantile")
prob_true_logit_12, prob_pred_logit_12 = calibration_curve(freq12["y_test"],␣
 ↪freq12["test_pred_logit"], n_bins=10, strategy="quantile")
prob_true_rf_12, prob_pred_rf_12 = calibration_curve(freq12["y_test"],␣
 ↪freq12["test_pred_rf"], n_bins=10, strategy="quantile")

plt.figure(figsize=(6, 6))
plt.plot([0, 0.25], [0, 0.25], linestyle="--", label="Perfect calibration")
plt.plot(prob_pred_lpm_12, prob_true_lpm_12, marker="o", label="LPM 12m")
plt.plot(prob_pred_logit_12, prob_true_logit_12, marker="o", label="Logit 12m")
plt.plot(prob_pred_rf_12, prob_true_rf_12, marker="o", label="RF 12m")
plt.xlabel("Mean predicted PD")
plt.ylabel("Observed default rate")
plt.title("12-month calibration on test cohort")
plt.legend()
plt.show()

# 18m calibration: LPM vs Logit vs RF
prob_true_lpm_18, prob_pred_lpm_18 = calibration_curve(freq18["y_test"],␣
 ↪freq18["test_pred_lpm"], n_bins=10, strategy="quantile")
prob_true_logit_18, prob_pred_logit_18 = calibration_curve(freq18["y_test"],␣
 ↪freq18["test_pred_logit"], n_bins=10, strategy="quantile")
prob_true_rf_18, prob_pred_rf_18 = calibration_curve(freq18["y_test"],␣
 ↪freq18["test_pred_rf"], n_bins=10, strategy="quantile")
```

```
plt.figure(figsize=(6, 6))
plt.plot([0, 0.32], [0, 0.32], linestyle="--", label="Perfect calibration")
plt.plot(prob_pred_lpm_18, prob_true_lpm_18, marker="o", label="LPM 18m")
plt.plot(prob_pred_logit_18, prob_true_logit_18, marker="o", label="Logit 18m")
plt.plot(prob_pred_rf_18, prob_true_rf_18, marker="o", label="RF 18m")
plt.xlabel("Mean predicted PD")
plt.ylabel("Observed default rate")
plt.title("18-month calibration on test cohort")
plt.legend()
plt.show()

# 24m calibration: compare all fixed-horizon models against Cox later
prob_true_logit_24, prob_pred_logit_24 = calibration_curve(freq24["y_test"],␣
 ↪freq24["test_pred_logit"], n_bins=10, strategy="quantile")
prob_true_rf_24, prob_pred_rf_24 = calibration_curve(freq24["y_test"],␣
 ↪freq24["test_pred_rf"], n_bins=10, strategy="quantile")
plt.figure(figsize=(6, 4))
plt.plot([0, 0.40], [0, 0.40], linestyle="--", label="Perfect calibration")
plt.plot(prob_pred_logit_24, prob_true_logit_24, marker="o", label="Logit 24m")
plt.plot(prob_pred_rf_24, prob_true_rf_24, marker="o", label="RF 24m")
plt.xlabel("Mean predicted PD")
plt.ylabel("Observed default rate")
plt.title("24-month calibration on test cohort")
plt.legend()
plt.show()
```


6-month calibration on test cohort
Perfect calibration
LPM 6m
Logit 6m
RF 6m
Observed default rate
Mean predicted PD
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175


12-month calibration on test cohort
Perfect calibration
LPM 12m
Logit 12m
RF 12m
Observed default rate
Mean predicted PD
0.00
0.05
0.10
0.15
0.20
0.25


18-month calibration on test cohort
Perfect calibration
LPM 18m
Logit 18m
RF 18m
Observed default rate
Mean predicted PD
0.00
0.05
0.10
0.15
0.20
0.25
0.30

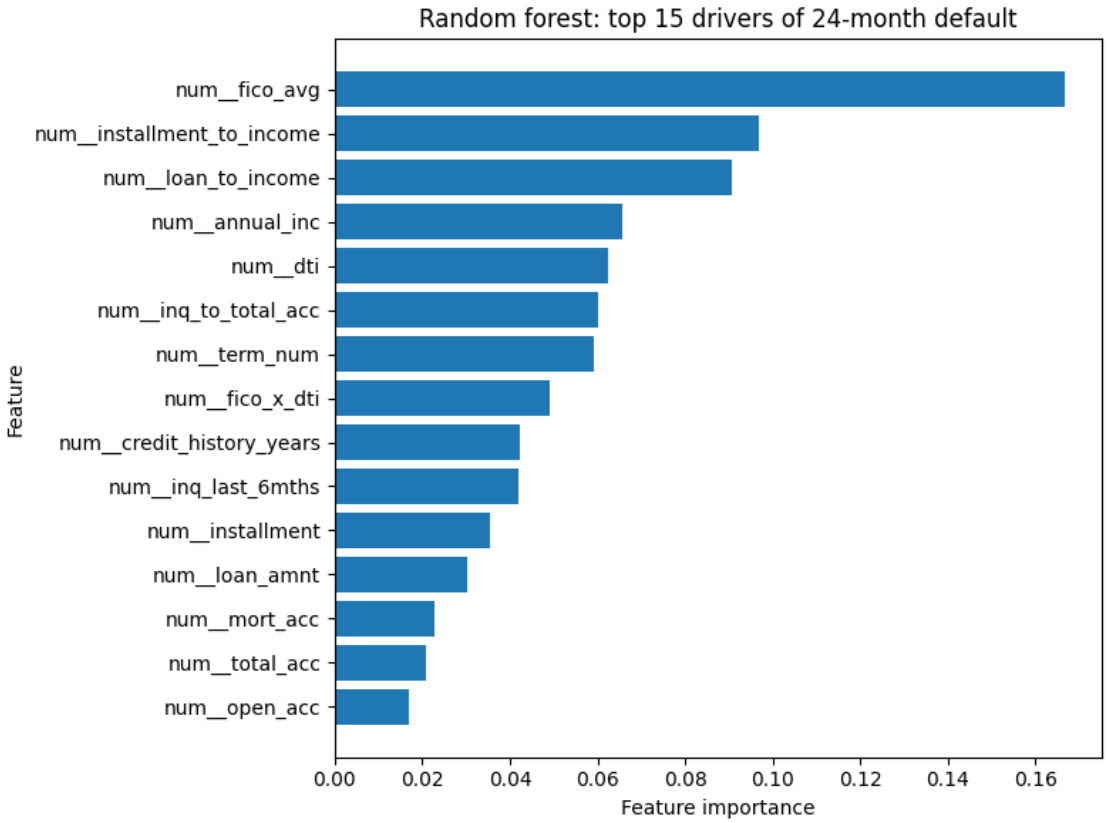


0.5.2 7.2 Random forest feature importance

The forest is a benchmark rather than the main PD model, so this view helps check whether it is picking up the same broad credit signals as the Cox fit.

[38]:
```
rf_feature_names_24 = freq24["rf_raw"].named_steps["prep"].
 ↪get_feature_names_out()
rf_feature_importance_24 = pd.DataFrame({
    "feature": rf_feature_names_24,
    "importance": freq24["rf_raw"].named_steps["model"].feature_importances_,
}).sort_values("importance", ascending=False)

display(rf_feature_importance_24.head(20))

plt.figure(figsize=(8, 6))
top_rf_importance = rf_feature_importance_24.head(15).sort_values("importance")
plt.barh(top_rf_importance["feature"], top_rf_importance["importance"])
plt.xlabel("Feature importance")
plt.ylabel("Feature")
plt.title("Random forest: top 15 drivers of 24-month default")
plt.tight_layout()
plt.show()
```

	feature	importance
11	num__fico_avg	0.166791
15	num__installment_to_income	0.096938
14	num__loan_to_income	0.090506
2	num__annual_inc	0.065743
3	num__dti	0.062365
16	num__inq_to_total_acc	0.060067
13	num__term_num	0.059054
18	num__fico_x_dti	0.049220
12	num__credit_history_years	0.042223
5	num__inq_last_6mths	0.042072
1	num__installment	0.035375
0	num__loan_amnt	0.030118
9	num__mort_acc	0.022701
8	num__total_acc	0.020739
6	num__open_acc	0.016856
70	cat__purpose_credit_card	0.016312
88	cat__home_ownership_RENT	0.015821
84	cat__home_ownership_MORTGAGE	0.015651
100	cat__verification_status_Not Verified	0.014093
80	cat__purpose_small_business	0.007908



0.6 8. Cox PH as the time-to-default extension

[39]:
```
cox_numeric = model_numeric.copy()
cox_categorical = model_categorical.copy()

def prepare_cox_design(train_df, eval_df):
    train_term = train_df[["id", "duration_months", "default_event"] +␣
 ↪cox_numeric + cox_categorical].copy()
    eval_term = eval_df[["id", "duration_months", "default_event"] +␣
 ↪cox_numeric + cox_categorical].copy()

    for frame in [train_term, eval_term]:
        for c in cox_numeric + ["duration_months", "default_event"]:
            frame[c] = pd.to_numeric(frame[c], errors="coerce")
        for c in cox_categorical:
            frame[c] = frame[c].astype("string").fillna("Missing")

    train_term = train_term.dropna(subset=["duration_months", "default_event"]).
 ↪copy()
    eval_term = eval_term.dropna(subset=["duration_months", "default_event"]).
 ↪copy()

    train_medians = train_term[cox_numeric].median(numeric_only=True)
    train_term[cox_numeric] = train_term[cox_numeric].fillna(train_medians)
    eval_term[cox_numeric] = eval_term[cox_numeric].fillna(train_medians)

    # Stabilize Cox optimization: cap heavy tails and standardize numeric inputs
    q_low = train_term[cox_numeric].quantile(0.01)
    q_high = train_term[cox_numeric].quantile(0.99)
    train_term[cox_numeric] = train_term[cox_numeric].clip(lower=q_low,␣
 ↪upper=q_high, axis=1)
    eval_term[cox_numeric] = eval_term[cox_numeric].clip(lower=q_low,␣
 ↪upper=q_high, axis=1)

    train_mean = train_term[cox_numeric].mean()
    train_std = train_term[cox_numeric].std().replace(0, 1)
    train_term[cox_numeric] = (train_term[cox_numeric] - train_mean) / train_std
    eval_term[cox_numeric] = (eval_term[cox_numeric] - train_mean) / train_std

    train_dm = pd.get_dummies(train_term, columns=cox_categorical,␣
 ↪drop_first=True)
    eval_dm = pd.get_dummies(eval_term, columns=cox_categorical,␣
 ↪drop_first=True)

    feature_cols = [c for c in train_dm.columns if c not in ["id",␣
 ↪"duration_months", "default_event"]]
```

```
    for c in feature_cols:
        if c not in eval_dm.columns:
            eval_dm[c] = 0

    extra_eval = [c for c in eval_dm.columns if c not in ["id",␣
 ↪"duration_months", "default_event"] + feature_cols]
    if extra_eval:
        eval_dm = eval_dm.drop(columns=extra_eval)

    train_dm = train_dm[["id", "duration_months", "default_event"] +␣
 ↪feature_cols].copy()
    eval_dm = eval_dm[["id", "duration_months", "default_event"] +␣
 ↪feature_cols].copy()

    for frame in [train_dm, eval_dm]:
        bool_cols = frame.select_dtypes(include=["bool"]).columns
        if len(bool_cols) > 0:
            frame[bool_cols] = frame[bool_cols].astype(int)
        frame[feature_cols] = frame[feature_cols].astype(float)
        frame["duration_months"] = frame["duration_months"].astype(float)
        frame["default_event"] = frame["default_event"].astype(int)

    nunique = train_dm[feature_cols].nunique(dropna=False)
    constant_cols = nunique[nunique <= 1].index.tolist()
    if constant_cols:
        feature_cols = [c for c in feature_cols if c not in constant_cols]
        train_dm = train_dm[["id", "duration_months", "default_event"] +␣
 ↪feature_cols].copy()
        eval_dm = eval_dm[["id", "duration_months", "default_event"] +␣
 ↪feature_cols].copy()

    return train_dm, eval_dm, feature_cols

cox_train, cox_valid, cox_features = prepare_cox_design(train_loans,␣
 ↪valid_loans)
_, cox_test, _ = prepare_cox_design(train_loans, test_loans)

cph = CoxPHFitter(penalizer=0.1)
cph.fit(
    cox_train.drop(columns=["id"]),
    duration_col="duration_months",
    event_col="default_event",
    show_progress=False,
)

valid_risk = cph.predict_partial_hazard(cox_valid[cox_features]).values.ravel()
```

```
test_risk = cph.predict_partial_hazard(cox_test[cox_features]).values.ravel()

cox_c_valid = concordance_index(
    cox_valid["duration_months"],
    -valid_risk,
    cox_valid["default_event"],
)
cox_c_test = concordance_index(
    cox_test["duration_months"],
    -test_risk,
    cox_test["default_event"],
)

cox_summary = pd.DataFrame([{
    "model": "Cox_PH_time_to_default",
    "valid_c_index": cox_c_valid,
    "test_c_index": cox_c_test,
    "n_train": len(cox_train),
    "n_valid": len(cox_valid),
    "n_test": len(cox_test),
}])

display(cox_summary)
```

```
                    model  valid_c_index  test_c_index  n_train  n_valid  \
0  Cox_PH_time_to_default       0.653339      0.655584   466345   421095

    n_test
0  1373228
```

0.6.1 8.1 The strongest Cox hazard ratios

[40]:
```
cox_coef_view = cph.summary[["coef", "exp(coef)", "se(coef)", "z", "p"]].
 ↪sort_values("exp(coef)", ascending=False)
display(cox_coef_view.head(20))
```

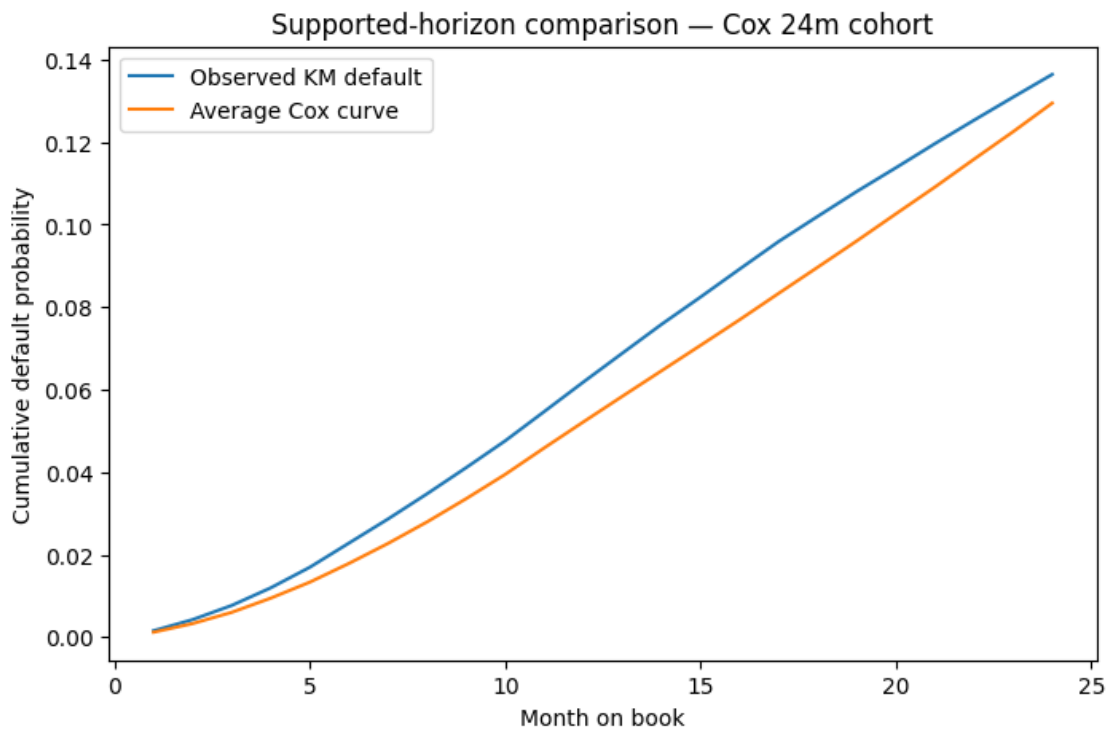
```
                             coef  exp(coef)  se(coef)          z  \
covariate
addr_state_NE            0.945249   2.573453  0.351965   2.685630
purpose_small_business   0.403724   1.497391  0.021673  18.627729
purpose_renewable_energy 0.238241   1.269014  0.103408   2.303888
purpose_moving           0.226980   1.254805  0.035241   6.440720
addr_state_IA            0.186640   1.205193  0.491914   0.379415
purpose_medical          0.184435   1.202539  0.029221   6.311812
purpose_educational      0.178093   1.194937  0.090561   1.966552
emp_length_Missing       0.154240   1.166771  0.013566  11.369866
home_ownership_OTHER     0.143452   1.154251  0.136826   1.048427
purpose_vacation         0.134939   1.144467  0.040451   3.335817
```

```
addr_state_NV             0.133345   1.142644  0.023713   5.623370
purpose_other             0.132044   1.141159  0.013462   9.808655
purpose_house             0.128802   1.137465  0.040843   3.153572
addr_state_MS             0.119665   1.127119  0.051250   2.334946
term_num                  0.107480   1.113469  0.003002  35.805712
addr_state_AL             0.089816   1.093973  0.024762   3.627153
inq_last_6mths            0.079532   1.082781  0.003383  23.508620
installment_to_income     0.075817   1.078766  0.003723  20.362257
addr_state_TN             0.071045   1.073629  0.024418   2.909516
home_ownership_RENT       0.065743   1.067952  0.007290   9.018472

                                        p
covariate
addr_state_NE              7.239312e-03
purpose_small_business     1.914785e-77
purpose_renewable_energy   2.122892e-02
purpose_moving             1.189084e-10
addr_state_IA              7.043799e-01
purpose_medical            2.757867e-10
purpose_educational        4.923489e-02
emp_length_Missing         5.907817e-30
home_ownership_OTHER       2.944421e-01
purpose_vacation           8.504905e-04
addr_state_NV              1.872673e-08
purpose_other              1.033362e-22
purpose_house              1.612854e-03
addr_state_MS              1.954622e-02
term_num                  9.000145e-281
addr_state_AL              2.865637e-04
inq_last_6mths            3.329461e-122
installment_to_income      3.615374e-92
addr_state_TN              3.619884e-03
home_ownership_RENT        1.907306e-19
```

0.6.2 8.2 Supported-horizon Cox curve check

[41]:
```
def km_curve(loans, max_horizon=None):
    kmf = KaplanMeierFitter()
    kmf.fit(
        durations=loans["duration_months"],
        event_observed=loans["default_event"],
    )
    surv = kmf.survival_function_.reset_index()
    surv.columns = ["month_on_book", "survival"]
    surv["month_on_book"] = surv["month_on_book"].astype(float).round().
 ↪astype(int)
    surv["cum_default_km"] = 1 - surv["survival"]
```

```
    if max_horizon is not None:
        surv = surv[surv["month_on_book"] <= max_horizon].copy()
    return surv

km_test = km_curve(test_loans, max_horizon=LONG_HORIZON)

surv_pred_test = cph.predict_survival_function(cox_test[cox_features])
avg_surv = surv_pred_test.mean(axis=1).reset_index()
avg_surv.columns = ["month_on_book", "survival_cox_avg"]
avg_surv["month_on_book"] = avg_surv["month_on_book"].astype(float).round().
 ↪astype(int)
avg_surv = avg_surv.groupby("month_on_book",␣
 ↪as_index=False)["survival_cox_avg"].last()
avg_surv["cum_default_cox_avg"] = 1 - avg_surv["survival_cox_avg"]

compare_curve = km_test.merge(
    avg_surv[["month_on_book", "cum_default_cox_avg"]],
    on="month_on_book",
    how="inner",
)

plt.figure(figsize=(8, 5))
plt.plot(compare_curve["month_on_book"], compare_curve["cum_default_km"],␣
 ↪label="Observed KM default")
plt.plot(compare_curve["month_on_book"], compare_curve["cum_default_cox_avg"],␣
 ↪label="Average Cox curve")
plt.xlabel("Month on book")
plt.ylabel("Cumulative default probability")
plt.title(f"Supported-horizon comparison - Cox {LONG_HORIZON}m cohort")
plt.legend()
plt.show()
```



0.7 9. Convert Cox into a simple 24-month PD and compare it with 24-month logit

[42]:
```
# 18m PD from Cox on the validation/test sets
cox_valid_18 = cph.predict_survival_function(cox_valid[cox_features],␣
 ↪times=[EIGHTEEN_HORIZON]).T
cox_valid_18.columns = ["survival_18m"]
cox_valid_18["pd_18m_cox"] = 1 - cox_valid_18["survival_18m"]
cox_valid_18 = cox_valid_18.reset_index(drop=True)
cox_valid_18["id"] = cox_valid["id"].values

cox_test_18 = cph.predict_survival_function(cox_test[cox_features],␣
 ↪times=[EIGHTEEN_HORIZON]).T
cox_test_18.columns = ["survival_18m"]
cox_test_18["pd_18m_cox"] = 1 - cox_test_18["survival_18m"]
cox_test_18 = cox_test_18.reset_index(drop=True)
cox_test_18["id"] = cox_test["id"].values

valid18_compare = valid_18[["id", f"default_within_{EIGHTEEN_HORIZON}m"]].merge(
    cox_valid_18[["id", "pd_18m_cox"]],
    on="id",
    how="inner",
)
```

```
test18_compare = test_18[["id", f"default_within_{EIGHTEEN_HORIZON}m"]].merge(
    cox_test_18[["id", "pd_18m_cox"]],
    on="id",
    how="inner",
)

valid18_compare["pd_18m_logit"] = freq18["valid_pred_logit"]
test18_compare["pd_18m_logit"] = freq18["test_pred_logit"]
valid18_compare["pd_18m_rf"] = freq18["valid_pred_rf"]
test18_compare["pd_18m_rf"] = freq18["test_pred_rf"]

y_valid_18 = valid18_compare[f"default_within_{EIGHTEEN_HORIZON}m"]
y_test_18 = test18_compare[f"default_within_{EIGHTEEN_HORIZON}m"]

# 24m PD from Cox on the validation/test sets
cox_valid_24 = cph.predict_survival_function(cox_valid[cox_features],␣
 ↪times=[LONG_HORIZON]).T
cox_valid_24.columns = ["survival_24m"]
cox_valid_24["pd_24m_cox"] = 1 - cox_valid_24["survival_24m"]
cox_valid_24 = cox_valid_24.reset_index(drop=True)
cox_valid_24["id"] = cox_valid["id"].values

cox_test_24 = cph.predict_survival_function(cox_test[cox_features],␣
 ↪times=[LONG_HORIZON]).T
cox_test_24.columns = ["survival_24m"]
cox_test_24["pd_24m_cox"] = 1 - cox_test_24["survival_24m"]
cox_test_24 = cox_test_24.reset_index(drop=True)
cox_test_24["id"] = cox_test["id"].values

valid24_compare = valid_24[["id", f"default_within_{LONG_HORIZON}m"]].merge(
    cox_valid_24[["id", "pd_24m_cox"]],
    on="id",
    how="inner",
)
test24_compare = test_24[["id", f"default_within_{LONG_HORIZON}m"]].merge(
    cox_test_24[["id", "pd_24m_cox"]],
    on="id",
    how="inner",
)

valid24_compare["pd_24m_logit"] = freq24["valid_pred_logit"]
test24_compare["pd_24m_logit"] = freq24["test_pred_logit"]
valid24_compare["pd_24m_rf"] = freq24["valid_pred_rf"]
test24_compare["pd_24m_rf"] = freq24["test_pred_rf"]

y_valid_24 = valid24_compare[f"default_within_{LONG_HORIZON}m"]
y_test_24 = test24_compare[f"default_within_{LONG_HORIZON}m"]
```

```
longterm_results = pd.DataFrame([
    {
        "model": "Logit_18m",
        "valid_roc_auc": roc_auc_score(y_valid_18,␣
↪valid18_compare["pd_18m_logit"]),
        "valid_pr_auc": average_precision_score(y_valid_18,␣
↪valid18_compare["pd_18m_logit"]),
        "valid_brier": brier_score_loss(y_valid_18,␣
↪valid18_compare["pd_18m_logit"]),
        "test_roc_auc": roc_auc_score(y_test_18,␣
↪test18_compare["pd_18m_logit"]),
        "test_pr_auc": average_precision_score(y_test_18,␣
↪test18_compare["pd_18m_logit"]),
        "test_brier": brier_score_loss(y_test_18,␣
↪test18_compare["pd_18m_logit"]),
    },
    {
        "model": "Cox_implied_PD_18m",
        "valid_roc_auc": roc_auc_score(y_valid_18,␣
↪valid18_compare["pd_18m_cox"]),
        "valid_pr_auc": average_precision_score(y_valid_18,␣
↪valid18_compare["pd_18m_cox"]),
        "valid_brier": brier_score_loss(y_valid_18,␣
↪valid18_compare["pd_18m_cox"]),
        "test_roc_auc": roc_auc_score(y_test_18, test18_compare["pd_18m_cox"]),
        "test_pr_auc": average_precision_score(y_test_18,␣
↪test18_compare["pd_18m_cox"]),
        "test_brier": brier_score_loss(y_test_18, test18_compare["pd_18m_cox"]),
    },
    {
        "model": "RF_18m",
        "valid_roc_auc": roc_auc_score(y_valid_18,␣
↪valid18_compare["pd_18m_rf"]),
        "valid_pr_auc": average_precision_score(y_valid_18,␣
↪valid18_compare["pd_18m_rf"]),
        "valid_brier": brier_score_loss(y_valid_18,␣
↪valid18_compare["pd_18m_rf"]),
        "test_roc_auc": roc_auc_score(y_test_18, test18_compare["pd_18m_rf"]),
        "test_pr_auc": average_precision_score(y_test_18,␣
↪test18_compare["pd_18m_rf"]),
        "test_brier": brier_score_loss(y_test_18, test18_compare["pd_18m_rf"]),
    },
    {
        "model": "Logit_24m",
```

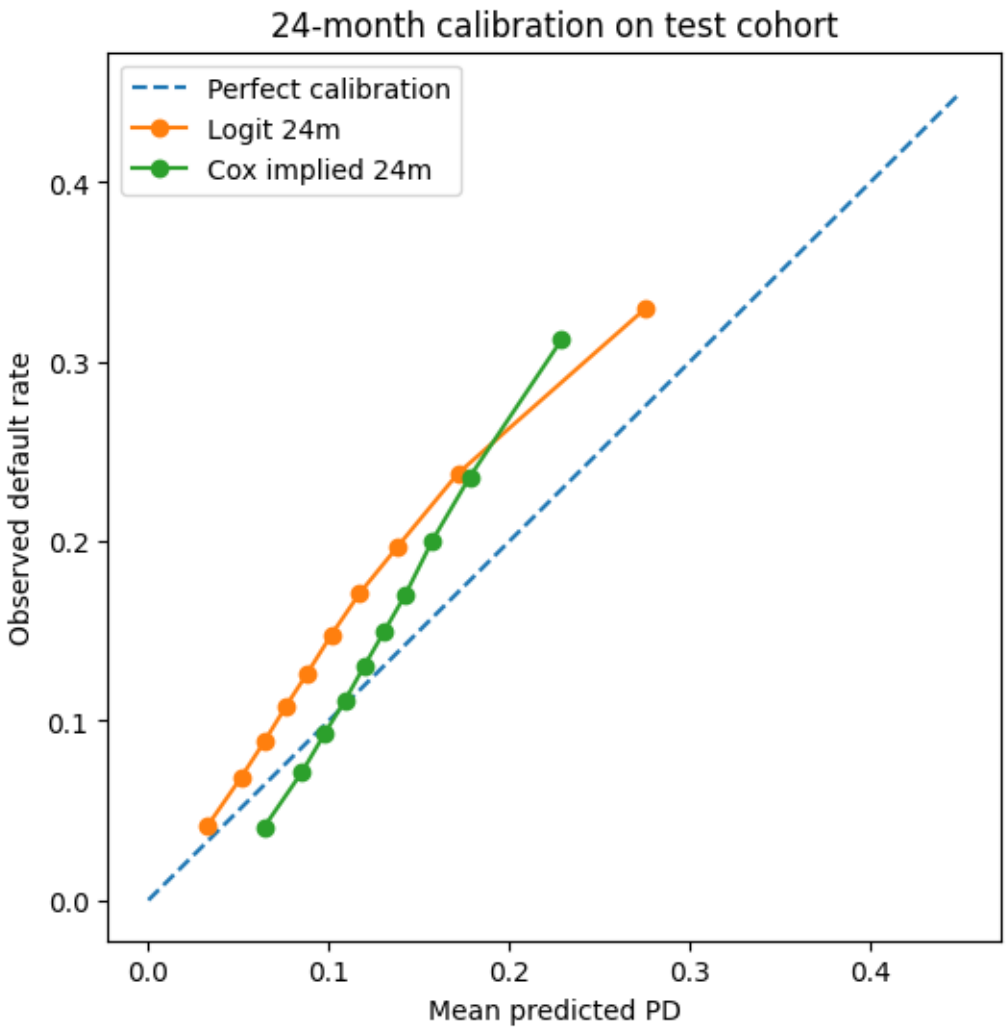
```
        "valid_roc_auc": roc_auc_score(y_valid_24,␣
↪valid24_compare["pd_24m_logit"]),
        "valid_pr_auc": average_precision_score(y_valid_24,␣
↪valid24_compare["pd_24m_logit"]),
        "valid_brier": brier_score_loss(y_valid_24,␣
↪valid24_compare["pd_24m_logit"]),
        "test_roc_auc": roc_auc_score(y_test_24,␣
↪test24_compare["pd_24m_logit"]),
        "test_pr_auc": average_precision_score(y_test_24,␣
↪test24_compare["pd_24m_logit"]),
        "test_brier": brier_score_loss(y_test_24,␣
↪test24_compare["pd_24m_logit"]),
    },
    {
        "model": "Cox_implied_PD_24m",
        "valid_roc_auc": roc_auc_score(y_valid_24,␣
↪valid24_compare["pd_24m_cox"]),
        "valid_pr_auc": average_precision_score(y_valid_24,␣
↪valid24_compare["pd_24m_cox"]),
        "valid_brier": brier_score_loss(y_valid_24,␣
↪valid24_compare["pd_24m_cox"]),
        "test_roc_auc": roc_auc_score(y_test_24, test24_compare["pd_24m_cox"]),
        "test_pr_auc": average_precision_score(y_test_24,␣
↪test24_compare["pd_24m_cox"]),
        "test_brier": brier_score_loss(y_test_24, test24_compare["pd_24m_cox"]),
    },
    {
        "model": "RF_24m",
        "valid_roc_auc": roc_auc_score(y_valid_24,␣
↪valid24_compare["pd_24m_rf"]),
        "valid_pr_auc": average_precision_score(y_valid_24,␣
↪valid24_compare["pd_24m_rf"]),
        "valid_brier": brier_score_loss(y_valid_24,␣
↪valid24_compare["pd_24m_rf"]),
        "test_roc_auc": roc_auc_score(y_test_24, test24_compare["pd_24m_rf"]),
        "test_pr_auc": average_precision_score(y_test_24,␣
↪test24_compare["pd_24m_rf"]),
        "test_brier": brier_score_loss(y_test_24, test24_compare["pd_24m_rf"]),
    },
])

display(longterm_results)
```

```
                model  valid_roc_auc  valid_pr_auc  valid_brier  test_roc_auc  \
0           Logit_18m       0.682818      0.187610     0.087569      0.675847
1  Cox_implied_PD_18m       0.670660      0.175989     0.088061      0.666398
2              RF_18m       0.670218      0.176071     0.087967      0.666832
```

```
3           Logit_24m       0.680637      0.238374     0.110807      0.679218
4  Cox_implied_PD_24m       0.670445      0.226868     0.111403      0.670676
5              RF_24m       0.669231      0.226967     0.111383      0.671190

   test_pr_auc  test_brier
0     0.198873    0.095529
1     0.188775    0.095234
2     0.190046    0.095451
3     0.266726    0.124262
4     0.254872    0.123877
5     0.256947    0.124861
```

0.7.1 9.1 Calibration: 24m logit vs 24m Cox

[43]:
```
prob_true_logit_24, prob_pred_logit_24 = calibration_curve(
    y_test_24, test24_compare["pd_24m_logit"], n_bins=10, strategy="quantile"
)
prob_true_cox_24, prob_pred_cox_24 = calibration_curve(
    y_test_24, test24_compare["pd_24m_cox"], n_bins=10, strategy="quantile"
)

plt.figure(figsize=(6, 6))
plt.plot([0, 0.45], [0, 0.45], linestyle="--", label="Perfect calibration")
plt.plot(prob_pred_logit_24, prob_true_logit_24, marker="o", label="Logit 24m")
plt.plot(prob_pred_cox_24, prob_true_cox_24, marker="o", label="Cox implied␣
 ↪24m")
plt.xlabel("Mean predicted PD")
plt.ylabel("Observed default rate")
plt.title("24-month calibration on test cohort")
plt.legend()
plt.show()
```



[44]:
```
# Calibrate Cox PDs at 24m, 36m, and 60m using isotonic regression.
calibration_horizons = [LONG_HORIZON, 36, 60]

if all(name in globals() for name in ["final_cph", "final_cox_all",␣
 ↪"final_cox_features"]):
    cox_model_for_cal = final_cph
    cox_design_for_cal = final_cox_all
    cox_features_for_cal = final_cox_features
else:
    cox_model_for_cal = cph
    cox_design_for_cal = pd.concat([cox_valid, cox_test], ignore_index=True)
    cox_features_for_cal = cox_features


def predict_cox_pd_for_ids(horizon, id_series):
```

```
    subset = cox_design_for_cal.loc[cox_design_for_cal["id"].isin(id_series),␣
↪["id"] + cox_features_for_cal].copy()
    pd_raw = 1 - cox_model_for_cal.predict_survival_function(
        subset[cox_features_for_cal],
        times=[horizon],
    ).T.iloc[:, 0].values
    return pd.DataFrame({"id": subset["id"].values, "pd_raw": pd_raw})


cox_isotonic_models = {}
cox_calibration_rows = []

for h in calibration_horizons:
    valid_h = make_horizon_label(valid_loans, h)
    test_h = make_horizon_label(test_loans, h)

    # Only calibrate horizon h on contracts that can actually run that long.
    valid_h = valid_h.loc[valid_h["term_num"] >= h, ["id",␣
↪f"default_within_{h}m"]].copy()
    test_h = test_h.loc[test_h["term_num"] >= h, ["id",␣
↪f"default_within_{h}m"]].copy()

    pred_valid_h = predict_cox_pd_for_ids(h, valid_h["id"])
    pred_test_h = predict_cox_pd_for_ids(h, test_h["id"])

    valid_cal_h = valid_h.merge(pred_valid_h, on="id", how="inner")
    test_cal_h = test_h.merge(pred_test_h, on="id", how="inner")

    y_valid_h = valid_cal_h[f"default_within_{h}m"].values
    y_test_h = test_cal_h[f"default_within_{h}m"].values
    pd_raw_valid_h = valid_cal_h["pd_raw"].values
    pd_raw_test_h = test_cal_h["pd_raw"].values

    iso_h = IsotonicRegression(out_of_bounds="clip")
    iso_h.fit(pd_raw_valid_h, y_valid_h)
    cox_isotonic_models[h] = iso_h

    cox_calibration_rows.append({
        "horizon_months": h,
        "model": f"Cox_implied_PD_{h}m_raw",
        "n_valid": len(valid_cal_h),
        "n_test": len(test_cal_h),
        "valid_mean_pd": pd_raw_valid_h.mean(),
        "test_mean_pd": pd_raw_test_h.mean(),
        "valid_brier": brier_score_loss(y_valid_h, pd_raw_valid_h),
        "test_brier": brier_score_loss(y_test_h, pd_raw_test_h),
    })
```

```
    cox_calibration_rows.append({
        "horizon_months": h,
        "model": f"Cox_implied_PD_{h}m_calibrated",
        "n_valid": len(valid_cal_h),
        "n_test": len(test_cal_h),
        "valid_mean_pd": iso_h.transform(pd_raw_valid_h).mean(),
        "test_mean_pd": iso_h.transform(pd_raw_test_h).mean(),
        "valid_brier": brier_score_loss(y_valid_h, iso_h.
 ↪transform(pd_raw_valid_h)),
        "test_brier": brier_score_loss(y_test_h, iso_h.
 ↪transform(pd_raw_test_h)),
    })

# Backward-compatibility for downstream cells that reference this name directly.
cox_24_isotonic = cox_isotonic_models[LONG_HORIZON]

cox_calibration_summary = pd.DataFrame(cox_calibration_rows)
display(cox_calibration_summary)
```

```
   horizon_months                          model  n_valid  n_test  \
0              24          Cox_implied_PD_24m_raw   421095  707053
1              24   Cox_implied_PD_24m_calibrated   421095  707053
2              36          Cox_implied_PD_36m_raw   421042  536971
3              36   Cox_implied_PD_36m_calibrated   421042  536971
4              60          Cox_implied_PD_60m_raw    92520  116560
5              60   Cox_implied_PD_60m_calibrated    92520  116560

   valid_mean_pd  test_mean_pd  valid_brier  test_brier
0       0.147013      0.141214     0.111089    0.122992
1       0.133633      0.125718     0.110488    0.122842
2       0.216442      0.209038     0.137777    0.158933
3       0.174173      0.165215     0.135681    0.160644
4       0.353605      0.344719     0.216119    0.207845
5       0.363943      0.351623     0.214815    0.207583
```

0.8 10. Choose the simplest long-term model

[45]:
```
lt = longterm_results.set_index("model")
median_24m_models = ["Logit_24m", "RF_24m", "Cox_implied_PD_24m"]

selection_table = longterm_results.copy()
selection_table["chosen_for_simple_longterm"] = False
display(selection_table)

chosen_longterm_model = "Median_24m_of_Logit_RF_Cox"
print("Chosen simple long-term PD model:", chosen_longterm_model)
```

```
                 model  valid_roc_auc  valid_pr_auc  valid_brier  test_roc_auc  \
```

```
0           Logit_18m       0.682818      0.187610     0.087569      0.675847
1  Cox_implied_PD_18m       0.670660      0.175989     0.088061      0.666398
2              RF_18m       0.670218      0.176071     0.087967      0.666832
3           Logit_24m       0.680637      0.238374     0.110807      0.679218
4  Cox_implied_PD_24m       0.670445      0.226868     0.111403      0.670676
5              RF_24m       0.669231      0.226967     0.111383      0.671190

   test_pr_auc  test_brier  chosen_for_simple_longterm
0     0.198873    0.095529                       False
1     0.188775    0.095234                       False
2     0.190046    0.095451                       False
3     0.266726    0.124262                       False
4     0.254872    0.123877                       False
5     0.256947    0.124861                       False

Chosen simple long-term PD model: Median_24m_of_Logit_RF_Cox
```

0.9 11. Refit final production-style models on train + validation

[46]:
```
# Combine train+valid for final fitting
tv_loans = pd.concat([train_loans, valid_loans], ignore_index=True)
tv_6 = pd.concat([train_6, valid_6], ignore_index=True)
tv_12 = pd.concat([train_12, valid_12], ignore_index=True)
tv_18 = pd.concat([train_18, valid_18], ignore_index=True)
tv_24 = pd.concat([train_24, valid_24], ignore_index=True)

# Final 6m logit
X_tv_6 = tv_6[model_numeric + model_categorical]
y_tv_6 = tv_6[f"default_within_{SIX_HORIZON}m"]

final_logit_6 = Pipeline([
    ("prep", preprocessor),
    ("model", LogisticRegression(max_iter=2000, C=0.5,␣
 ↪random_state=RANDOM_STATE)),
])
final_logit_6.fit(X_tv_6, y_tv_6)

# Final 18m logit
X_tv_18 = tv_18[model_numeric + model_categorical]
y_tv_18 = tv_18[f"default_within_{EIGHTEEN_HORIZON}m"]

final_logit_18 = Pipeline([
    ("prep", preprocessor),
    ("model", LogisticRegression(max_iter=2000, C=0.5,␣
 ↪random_state=RANDOM_STATE)),
])
final_logit_18.fit(X_tv_18, y_tv_18)
```

```
# Final 12m logit
X_tv_12 = tv_12[model_numeric + model_categorical]
y_tv_12 = tv_12[f"default_within_{SHORT_HORIZON}m"]

final_logit_12 = Pipeline([
    ("prep", preprocessor),
    ("model", LogisticRegression(max_iter=2000, C=0.5,␣
 ↪random_state=RANDOM_STATE)),
])
final_logit_12.fit(X_tv_12, y_tv_12)

# Final 24m logit
X_tv_24 = tv_24[model_numeric + model_categorical]
y_tv_24 = tv_24[f"default_within_{LONG_HORIZON}m"]

final_logit_24 = Pipeline([
    ("prep", preprocessor),
    ("model", LogisticRegression(max_iter=2000, C=0.5,␣
 ↪random_state=RANDOM_STATE)),
])
final_logit_24.fit(X_tv_24, y_tv_24)

# Final RF models for 6m and 18m
final_rf_raw_6 = Pipeline([
    ("prep", preprocessor),
    ("model", RandomForestClassifier(
        n_estimators=200,
        max_depth=12,
        min_samples_leaf=50,
        max_features="sqrt",
        class_weight="balanced_subsample",
        max_samples=0.7,
        random_state=RANDOM_STATE,
        n_jobs=-1,
    )),
])
final_rf_6 = CalibratedClassifierCV(estimator=final_rf_raw_6, method="sigmoid",␣
 ↪cv=3)
final_rf_6.fit(X_tv_6, y_tv_6)

final_rf_raw_18 = Pipeline([
    ("prep", preprocessor),
    ("model", RandomForestClassifier(
        n_estimators=200,
        max_depth=12,
        min_samples_leaf=50,
        max_features="sqrt",
```

```
        class_weight="balanced_subsample",
        max_samples=0.7,
        random_state=RANDOM_STATE,
        n_jobs=-1,
    )),
])
final_rf_18 = CalibratedClassifierCV(estimator=final_rf_raw_18,␣
 ↪method="sigmoid", cv=3)
final_rf_18.fit(X_tv_18, y_tv_18)

# Final RF models
final_rf_raw_12 = Pipeline([
    ("prep", preprocessor),
    ("model", RandomForestClassifier(
        n_estimators=200,
        max_depth=12,
        min_samples_leaf=50,
        max_features="sqrt",
        class_weight="balanced_subsample",
        max_samples=0.7,
        random_state=RANDOM_STATE,
        n_jobs=-1,
    )),
])
final_rf_12 = CalibratedClassifierCV(estimator=final_rf_raw_12,␣
 ↪method="sigmoid", cv=3)
final_rf_12.fit(X_tv_12, y_tv_12)

final_rf_raw_24 = Pipeline([
    ("prep", preprocessor),
    ("model", RandomForestClassifier(
        n_estimators=200,
        max_depth=12,
        min_samples_leaf=50,
        max_features="sqrt",
        class_weight="balanced_subsample",
        max_samples=0.7,
        random_state=RANDOM_STATE,
        n_jobs=-1,
    )),
])
final_rf_24 = CalibratedClassifierCV(estimator=final_rf_raw_24,␣
 ↪method="sigmoid", cv=3)
final_rf_24.fit(X_tv_24, y_tv_24)

# Final Cox
```

```
final_cox_train, final_cox_all, final_cox_features =␣
 ↪prepare_cox_design(tv_loans, df_model)
final_cph = CoxPHFitter(penalizer=0.1)
final_cph.fit(
    final_cox_train.drop(columns=["id"]),
    duration_col="duration_months",
    event_col="default_event",
    show_progress=False,
)
```

[46]: <lifelines.CoxPHFitter: fitted with 887440 total observations, 734375 right-censored observations>

0.10 12. Score the whole modeled portfolio

[51]:
```
pd_output = df_model[[
    "id",
    "grade",
    "issue_d",
    "loan_status",
    "loan_amnt",
    "int_rate",
    "annual_inc",
    "dti",
    "fico_avg",
    "term_num",
    "addr_state",
    "purpose",
    "home_ownership",
    "installment",
    "emp_length",
]].copy()

X_all = df_model[model_numeric + model_categorical]
pd_output["pd_6m_logit"] = final_logit_6.predict_proba(X_all)[:, 1]
pd_output["pd_12m_logit"] = final_logit_12.predict_proba(X_all)[:, 1]
pd_output["pd_18m_logit"] = final_logit_18.predict_proba(X_all)[:, 1]
pd_output["pd_24m_logit"] = final_logit_24.predict_proba(X_all)[:, 1]
pd_output["pd_6m_rf"] = final_rf_6.predict_proba(X_all)[:, 1]
pd_output["pd_12m_rf"] = final_rf_12.predict_proba(X_all)[:, 1]
pd_output["pd_18m_rf"] = final_rf_18.predict_proba(X_all)[:, 1]
pd_output["pd_24m_rf"] = final_rf_24.predict_proba(X_all)[:, 1]

# Cox implied PDs: use a direction-corrected and numerically stable mapping
cox_horizons = [SIX_HORIZON, EIGHTEEN_HORIZON, LONG_HORIZON, 36, 60]
lp_all = final_cph.
 ↪predict_log_partial_hazard(final_cox_all[final_cox_features]).values.ravel()
```

```
rel_risk = np.exp(np.clip(lp_all, -50, 50))
base_ch = final_cph.baseline_cumulative_hazard_.iloc[:, 0]
base_idx = base_ch.index.values.astype(float)
base_vals = base_ch.values.astype(float)

cox_all = pd.DataFrame({"id": final_cox_all["id"].values})
for h in cox_horizons:
    h0 = float(np.interp(float(h), base_idx, base_vals))
    surv_h = np.exp(-h0 * rel_risk)
    cox_all[f"pd_{h}m_cox_raw"] = 1 - surv_h

# Apply isotonic calibrators where available.
for h in [LONG_HORIZON, 36, 60]:
    raw_col = f"pd_{h}m_cox_raw"
    cal_col = f"pd_{h}m_cox"
    if "cox_isotonic_models" in globals() and h in cox_isotonic_models:
        cox_all[cal_col] = cox_isotonic_models[h].transform(cox_all[raw_col])
    else:
        cox_all[cal_col] = cox_all[raw_col]

pd_output = pd_output.merge(
    cox_all[[
        "id",
        "pd_6m_cox_raw",
        "pd_18m_cox_raw",
        "pd_24m_cox_raw",
        "pd_24m_cox",
        "pd_36m_cox_raw",
        "pd_36m_cox",
        "pd_60m_cox_raw",
        "pd_60m_cox",
    ]],
    on="id",
    how="left",
)

# Do not report horizon PDs for contracts shorter than that horizon.
pd_output.loc[pd_output["term_num"] < 36, ["pd_36m_cox_raw", "pd_36m_cox"]] =␣
 ↪np.nan
pd_output.loc[pd_output["term_num"] < 60, ["pd_60m_cox_raw", "pd_60m_cox"]] =␣
 ↪np.nan

pd_output["pd_24m_chosen"] = pd_output[["pd_24m_logit", "pd_24m_rf",␣
 ↪"pd_24m_cox"]].median(axis=1)
#pd_output["chosen_longterm_model"] = chosen_longterm_model

display(pd_output.head())
```

```
pd_output.to_csv(OUTPUT_DIR / "pd_predictions.csv", index=False)
frequency_results.to_csv(OUTPUT_DIR / "frequency_model_metrics.csv",␣
 ↪index=False)
longterm_results.to_csv(OUTPUT_DIR / "longterm_model_comparison.csv",␣
 ↪index=False)
cox_summary.to_csv(OUTPUT_DIR / "cox_time_to_default_summary.csv", index=False)

print("Saved:")
print("-", OUTPUT_DIR / "pd_predictions.csv")
print("-", OUTPUT_DIR / "frequency_model_metrics.csv")
print("-", OUTPUT_DIR / "longterm_model_comparison.csv")
print("-", OUTPUT_DIR / "cox_time_to_default_summary.csv")
```

```
         id grade    issue_d loan_status  loan_amnt  int_rate  annual_inc  \
0  68407277     C 2015-12-01  Fully Paid     3600.0     13.99     55000.0
1  68355089     C 2015-12-01  Fully Paid    24700.0     11.99     65000.0
2  68341763     B 2015-12-01  Fully Paid    20000.0     10.78     63000.0
3  66310712     C 2015-12-01     Current    35000.0     14.85    110000.0
4  68476807     F 2015-12-01  Fully Paid    10400.0     22.45    104433.0

     dti  fico_avg  term_num  … pd_24m_rf pd_6m_cox_raw pd_18m_cox_raw  \
0   5.91     677.0      36.0  …  0.078027      0.013362       0.070474
1  16.06     717.0      36.0  …  0.173926      0.038372       0.191494
2  10.78     697.0      60.0  …  0.087730      0.013240       0.069849
3  17.06     787.0      60.0  …  0.104948      0.013285       0.070080
4  25.37     697.0      60.0  …  0.135153      0.022711       0.117330

   pd_24m_cox_raw pd_24m_cox  pd_36m_cox_raw  pd_36m_cox  pd_60m_cox_raw  \
0        0.101131   0.063890        0.151817    0.090164             NaN
1        0.266641   0.291310        0.380563    0.371731             NaN
2        0.100250   0.063312        0.150533    0.085685        0.216771
3        0.100576   0.063312        0.151007    0.085685        0.217426
4        0.166465   0.161109        0.245123    0.208251        0.343694

   pd_60m_cox  pd_24m_chosen
0         NaN       0.068016
1         NaN       0.228257
2    0.170025       0.063312
3    0.170025       0.063312
4    0.353398       0.161109

[5 rows x 32 columns]
```

```
Saved:
- data/simplification_PD/pd_predictions.csv
- data/simplification_PD/frequency_model_metrics.csv
- data/simplification_PD/longterm_model_comparison.csv
```

- data/simplification_PD/cox_time_to_default_summary.csv

0.10.1 12.1 Sanity check: largest Cox PDs among fully paid loans

[52]:
```
fully_paid_high_cox = (
    pd_output.loc[
        pd_output["loan_status"].astype(str).str.contains("Fully Paid",␣
 ↪case=False, regex=False)
        & (pd_output["term_num"] >= 60)
    ]
    .sort_values("pd_60m_cox", ascending=False)
    .loc[:, [
        "id",
        "issue_d",
        "loan_status",
        "term_num",
        "loan_amnt",
        "annual_inc",
        "dti",
        "fico_avg",
        "pd_24m_logit",
        "pd_24m_rf",
        "pd_24m_cox",
        "pd_24m_chosen",
        "pd_60m_cox_raw",
        "pd_60m_cox",
    ]]
    .head(20)
)

display(fully_paid_high_cox)
```

```
                id    issue_d loan_status  term_num  loan_amnt  annual_inc  \
1773557    8694572 2013-11-01  Fully Paid      60.0    23500.0     65000.0
477958   128443257 2018-02-01  Fully Paid      60.0    21000.0     33000.0
2126904  121807012 2017-10-01  Fully Paid      60.0    30000.0      1000.0
2048414  126052429 2017-12-01  Fully Paid      60.0    20000.0      1000.0
2068308  125104322 2017-12-01  Fully Paid      60.0    15000.0     13200.0
944406   107089616 2017-05-01  Fully Paid      60.0    18000.0     15120.0
1037530   73628560 2016-03-01  Fully Paid      60.0    20000.0     55000.0
327874    43945242 2015-04-01  Fully Paid      60.0    24650.0     50000.0
1635369     613444 2010-11-01  Fully Paid      60.0     9250.0     19200.0
2147102  120592257 2017-10-01  Fully Paid      60.0    35000.0     80000.0
905781   110282370 2017-06-01  Fully Paid      60.0    16000.0     50000.0
443492   129896709 2018-03-01  Fully Paid      60.0    20000.0     17000.0
1198502   21811292 2014-09-01  Fully Paid      60.0    17250.0     48000.0
1321035   12487177 2014-03-01  Fully Paid      60.0    17250.0     48000.0
2124748  121775551 2017-10-01  Fully Paid      60.0    36000.0     50000.0
```

1280823	15739832	2014-05-01	Fully Paid	60.0	10000.0	39000.0
39318	66035498	2015-12-01	Fully Paid	60.0	11900.0	36000.0
1278636	16081143	2014-05-01	Fully Paid	60.0	10450.0	31000.0
570051	117229890	2017-08-01	Fully Paid	60.0	24000.0	30000.0
64907	65128915	2015-11-01	Fully Paid	60.0	10000.0	28800.0

	dti	fico_avg	pd_24m_logit	pd_24m_rf	pd_24m_cox	pd_24m_chosen	\
1773557	29.32	697.0	0.630770	0.354235	0.520000	0.520000	
477958	45.20	667.0	0.424231	0.332486	0.520000	0.424231	
2126904	324.73	672.0	1.000000	0.326976	0.520000	0.520000	
2048414	277.11	672.0	1.000000	0.304575	0.563897	0.563897	
2068308	141.82	667.0	0.909680	0.390126	0.520000	0.520000	
944406	50.87	662.0	0.858159	0.382987	0.520000	0.520000	
1037530	43.53	707.0	0.463091	0.323763	0.520000	0.463091	
327874	38.14	667.0	0.469290	0.311977	0.520000	0.469290	
1635369	8.50	667.0	0.450434	0.271689	0.520000	0.450434	
2147102	19.56	662.0	0.658256	0.340531	0.520000	0.520000	
905781	30.80	667.0	0.439780	0.351380	0.520000	0.439780	
443492	62.85	662.0	0.592145	0.341539	0.520000	0.520000	
1198502	27.28	667.0	0.497217	0.359048	0.520000	0.497217	
1321035	34.78	662.0	0.531702	0.342961	0.520000	0.520000	
2124748	30.24	682.0	0.279528	0.341250	0.520000	0.341250	
1280823	24.83	687.0	0.523579	0.312575	0.520000	0.520000	
39318	37.00	662.0	0.531251	0.319965	0.520000	0.520000	
1278636	27.53	662.0	0.493808	0.336904	0.520000	0.493808	
570051	87.28	662.0	0.855021	0.340453	0.520000	0.520000	
64907	28.92	667.0	0.422924	0.354739	0.520000	0.422924	

	pd_60m_cox_raw	pd_60m_cox
1773557	0.800740	1.000000
477958	0.806471	1.000000
2126904	0.807381	1.000000
2048414	0.817671	1.000000
2068308	0.815779	1.000000
944406	0.810755	1.000000
1037530	0.792979	0.924662
327874	0.749146	0.774194
1635369	0.743827	0.774194
2147102	0.772472	0.774194
905781	0.743226	0.774194
443492	0.742568	0.774194
1198502	0.737832	0.774194
1321035	0.734182	0.774194
2124748	0.741864	0.774194
1280823	0.736027	0.774194
39318	0.777091	0.774194
1278636	0.745729	0.774194
570051	0.782536	0.774194

```
64907          0.775107    0.774194
```

[54]:
```python
# ============================================================
# Save condensed final output: about 10,000 sporadically chosen rows
# ============================================================

CONDENSED_N = 10_000
condensed_dir = OUTPUT_DIR / "condensed_outputs"
condensed_dir.mkdir(parents=True, exist_ok=True)

# Choose columns useful for teammates / final paper review
condensed_cols = [
    "id",
    "grade",
    "issue_d",
    "loan_status",
    "loan_amnt",
    "int_rate",
    "annual_inc",
    "dti",
    "fico_avg",
    "term_num",
    "addr_state",
    "purpose",
    "home_ownership",
    "installment",
    "emp_length",

    # Main fixed-horizon PDs
    "pd_6m_logit",
    "pd_12m_logit",
    "pd_18m_logit",
    "pd_24m_logit",

    # 24m model comparison
    "pd_24m_rf",
    "pd_24m_cox_raw",
    "pd_24m_cox",
    "pd_24m_chosen",

    # Lifetime / Cox extension
    "pd_36m_cox_raw",
    "pd_36m_cox",
    "pd_60m_cox_raw",
    "pd_60m_cox",
    "pd_lifetime_cox",
    "chosen_longterm_model",
```

```
]

# Keep only columns that actually exist
condensed_cols = [c for c in condensed_cols if c in pd_output.columns]

# Sporadic/random sample across the whole output
condensed_random = pd_output[condensed_cols].sample(
    n=min(CONDENSED_N, len(pd_output)),
    random_state=RANDOM_STATE,
).sort_values("id")

condensed_random_path = condensed_dir / "pd_output_condensed_10000_random.csv"
condensed_random.to_csv(condensed_random_path, index=False)

print("Saved random condensed output to:")
print(condensed_random_path)
display(condensed_random.head(20))
```

```
Saved random condensed output to:
data/simplification_PD/condensed_outputs/pd_output_condensed_10000_random.csv

                id grade    issue_d  loan_status  loan_amnt  int_rate  \
1616451    1000030     A 2011-10-01   Fully Paid    20000.0      8.90
1673422  100003912     C 2017-03-01      Current    28500.0     13.99
1661116  100006518     B 2017-03-01   Fully Paid     6000.0     11.44
1674099  100007026     C 2017-03-01      Current     4200.0     12.74
958444   100013479     C 2017-04-01      Current     8000.0     14.99
1682755  100015263     D 2017-03-01      Current    16000.0     19.99
1682661  100026535     B 2017-03-01      Current    12000.0     11.49
1677320  100035491     B 2017-03-01      Current     4000.0     11.39
1681117  100036374     C 2017-03-01   Fully Paid     3925.0     15.99
1655888  100037421     D 2017-03-01   Fully Paid     4500.0     18.99
1684805  100054463     B 2017-03-01      Current     6400.0     11.39
1675264  100066604     C 2017-03-01      Current    11625.0     15.99
1689041  100071682     C 2017-03-01   Fully Paid     4000.0     12.74
1659689  100082965     A 2017-03-01      Current    15000.0      5.32
1678695  100097311     A 2017-03-01   Fully Paid     6000.0      7.24
1691294  100111198     B 2017-02-01   Fully Paid     5000.0      8.24
1657479  100115345     B 2017-03-01   Fully Paid     8000.0     11.49
1673662  100137983     B 2017-03-01   Fully Paid     6600.0     11.49
1687900  100138906     A 2017-03-01  Charged Off    16000.0      6.99
1673093  100146014     C 2017-03-01      Current    21000.0     14.99

         annual_inc    dti  fico_avg  term_num  … pd_18m_logit pd_24m_logit  \
1616451     72000.0   7.80     742.0      36.0  …     0.097519     0.120343
1673422     80010.0  16.65     722.0      60.0  …     0.043019     0.064070
1661116     37389.0   9.66     662.0      36.0  …     0.090467     0.118894
1674099     91000.0  17.82     672.0      36.0  …     0.059838     0.084099
```

958444	67000.0	14.22	667.0	36.0	…	0.086461	0.120438
1682755	68000.0	16.24	667.0	36.0	…	0.143219	0.187270
1682661	80000.0	20.34	697.0	60.0	…	0.106557	0.140671
1677320	52000.0	28.87	717.0	36.0	…	0.034725	0.051092
1681117	39500.0	14.31	687.0	36.0	…	0.062146	0.084671
1655888	15000.0	11.28	667.0	36.0	…	0.161849	0.212150
1684805	35000.0	5.32	727.0	36.0	…	0.052197	0.072694
1675264	52000.0	8.68	667.0	60.0	…	0.159561	0.219254
1689041	120000.0	19.47	697.0	36.0	…	0.054978	0.078376
1659689	300000.0	11.62	737.0	36.0	…	0.020426	0.030303
1678695	87000.0	11.66	682.0	36.0	…	0.057774	0.084276
1691294	66000.0	13.82	707.0	36.0	…	0.028534	0.041315
1657479	25716.0	7.75	667.0	36.0	…	0.049327	0.077257
1673662	28000.0	19.59	677.0	36.0	…	0.085319	0.118733
1687900	75000.0	11.47	757.0	36.0	…	0.017955	0.025935
1673093	90000.0	18.91	682.0	36.0	…	0.088061	0.118653

	pd_24m_rf	pd_24m_cox_raw	pd_24m_cox	pd_24m_chosen	pd_36m_cox_raw	\
1616451	0.122748	0.177193	0.180153	0.122748	0.260075	
1673422	0.072813	0.135995	0.119053	0.072813	0.202084	
1661116	0.114921	0.143610	0.127690	0.118894	0.212919	
1674099	0.086076	0.110001	0.080942	0.084099	0.164708	
958444	0.091149	0.152427	0.140656	0.120438	0.225399	
1682755	0.135177	0.165627	0.161109	0.161109	0.243950	
1682661	0.120067	0.151711	0.140656	0.140656	0.224389	
1677320	0.054139	0.074444	0.038705	0.051092	0.112613	
1681117	0.090110	0.110226	0.080942	0.084671	0.165034	
1655888	0.205357	0.221660	0.237760	0.212150	0.320915	
1684805	0.067047	0.102732	0.074564	0.072694	0.154149	
1675264	0.175361	0.190024	0.189549	0.189549	0.277819	
1689041	0.080887	0.105044	0.074975	0.078376	0.157512	
1659689	0.027962	0.057285	0.018026	0.027962	0.087079	
1678695	0.072778	0.127957	0.108330	0.084276	0.190591	
1691294	0.039076	0.077802	0.041700	0.041315	0.117580	
1657479	0.110916	0.119859	0.093570	0.093570	0.178953	
1673662	0.149392	0.148400	0.132412	0.132412	0.219709	
1687900	0.026886	0.065095	0.027601	0.026886	0.098732	
1673093	0.107477	0.128843	0.108330	0.108330	0.191861	

	pd_36m_cox	pd_60m_cox_raw	pd_60m_cox
1616451	0.234454	NaN	NaN
1673422	0.151195	0.286868	0.278546
1661116	0.167171	NaN	NaN
1674099	0.112362	NaN	NaN
958444	0.183170	NaN	NaN
1682755	0.208251	NaN	NaN
1682661	0.183170	0.316514	0.321447
1677320	0.054311	NaN	NaN

```
1681117    0.112362             NaN         NaN
1655888    0.300943             NaN         NaN
1684805    0.096980             NaN         NaN
1675264    0.246818        0.385804    0.411193
1689041    0.101980             NaN         NaN
1659689    0.024505             NaN         NaN
1678695    0.140342             NaN         NaN
1691294    0.056843             NaN         NaN
1657479    0.126258             NaN         NaN
1673662    0.176956             NaN         NaN
1687900    0.036902             NaN         NaN
1673093    0.140342             NaN         NaN

[20 rows x 27 columns]
```